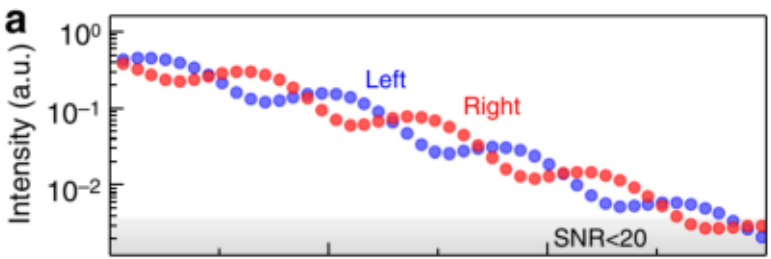
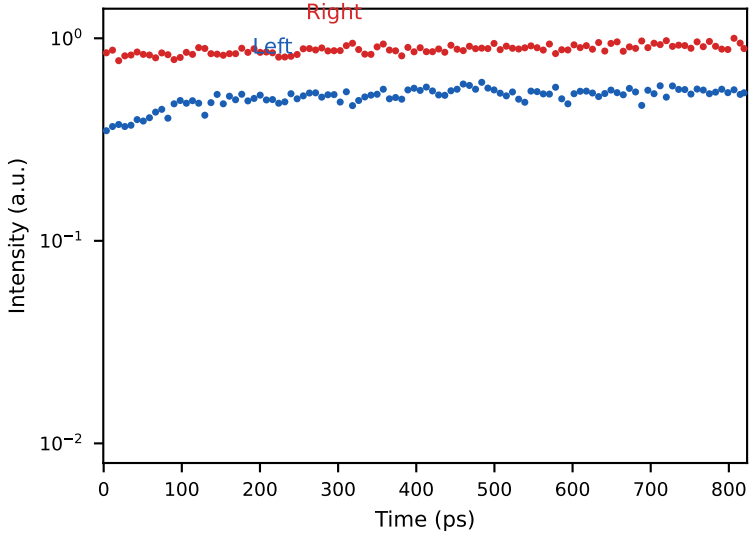
Adiyatullin et al. 2017 (Nat. Commun. 8:1329), Fig. 2(a)



This work: 26-08-04 HBT, 20000 frames, CFD50-timed



Left: original Fig. 2(a) of Adiyatullin et al. 2017 (Nat. Commun. 8:1329)— accumulated emission intensity of the left (blue) and right (red) mesa on a logarithmic scale, showing anti-phase population oscillations; the gray band marks the low-confidence SNR<20 region. Right: this work — replication on the 26-08-04 HBT batch (20000 single-sweep frames; photon-event source, rebin 8≈ 7.9 ps/bin, peak-normalized). The per-frame time zero is calibrated by CFD50, leading-edge constant-fraction discrimination on the reference pulse (50%-of-peak crossing, sub-row interpolated), placing the origin ~2.7 ps earlier than the former half-max centroid definition; the measured StreakCamBackG background is subtracted. Outcome: no anti-phase oscillation— cross-arm z fluctuations are uncorrelated ($r=0.03$–0.10, $p>0.14$); the 95% upper limit on a phase-locked sinusoidal z amplitude is 0.012–0.019 for 3–200 ps periods (~1–2% intensity modulation); the $t\leq 150$ ps left-side dip is an overdamped monotonic recovery, not an oscillation.